Data Format

Hi Vytis,
The excel sheet assigns the pulsing direction to the name of the scan. The scan is always taken after the pulse was applied, so a is as is, b is after a pulse along A, c is after a pulse along B, c2 is after 3 more pulses along B, d is after a pulse along A. They are in chronological and consecutive order, so we applied a pulse, did a scan, applied a different pulse, did a scan and so it. It is all the same sample that we applied more and more pulses to.

Best,
Max

Chapter 7 Thesis

Current-induced Metastable Sta7te in 1T-TaS2: Spatial Analysis
After having characterized the electronic structure of the current-induced metastable state in the previous chapter, this chapter focuses on its spatial evolution as a function of pulsing direction and number of pulses. The changes in the electronic structure on the sample surface are then put in context with the real-space structure obtained from microscope images and with changes in the bulk transport behavior.
7.1. Introduction to Spatial Mapping
The small spotsize of the nanoARPES endstation (~1 μm diameter) allows to scan the surface and analyze the changes in electronic structure in a spatially resolved manner. Figure 7.1 introduces the concept and demonstrates the wealth of information that can be gained by this technique. Figure 7.1a shows a microscope image of a cleaved sam- ple taken after the experiment, revealing various different domains on the surface. Fig- ure 7.1b shows a map of the energy and angle-integrated photoemission intensity of the same sample. For better clarity, regions with negligible intensity are omitted. Here, char- acteristic steps between different surface planes from the microscope image are clearly reflected in the data as pointed out by the blue arrows. In Figure 7.1c, the photoemission map is plotted directly on top of the optical microscope image, revealing an excellent cor- relation of all characteristic features. In particular, the outline of the cleaved area of the device is precisely reproduced by the map taken with ARPES. This establishes a direct connection between the two imaging methods and allows to correlate the spatial resolved changes in the electronic band structure after current pulsing with the real space surface structure.
The real space inhomogeneity is partially reflected in the ARPES spectra, as different planes on the surface can have slightly different orientations. Figure 7.1d-f show spec- tra extracted from three different regions (integration regions shown in panel b). They show clear differences both in overall shape and center of the bands, indicating that the measured cuts are shifted along both kx and ky with respect to each other. This suggests that the domains are not perfectly flat but instead to different degrees slightly tilted away from the surface. Figure 7.1g shows angle-integrated EDCs corresponding to the spectra in panels d-f, each showing three pronounced peaks (labeled as p1-p3). For clarity, EDCs are normalized to the same height. By comparison with Figure 6.3 p1 can be assigned to the LHB, p2 to the B1 band and p3 to the B2

band, respectively. As can be seen in the EDCs, the p1 position is almost identical for the different domain orientations, whereas p3 shows the clearest differences in its peak center. All angle-integrated EDCs can be well fitted with three peaks of Gaussian lineshape, corresponding to p1-p3. Figure 7.1h shows a spatial map based on the fitted position of p1. By comparison with Figure 7.1g, this gives mostly an overview of electronic structure

Figure 7.1.: a) Microscope image of a cleaved device taken after the experiment. b) Map of the integrated photoemission intensity of the same device. c) Photoemis- sion map from panel b superimposed on the optical microscope image from panel a. d-f) APRES spectra extracted from different regions of the device sur- face. Integration regions are indicated in panel b. g) Angle integrated curves corresponding to the spectra shown in panels d-f. Three clear peaks (p1-p3) can be identified. h) Spatial map of the p1 position obtained by fitting angle integrated curves. i) Spatial map of the p3 position. j) Spatial map of Irat.

with only minor contributions of spatial inhomogeneities. Indeed, the map has a differ- ent topography than the integrated photoemission intensity in Figure 7.1b, most notably showing a small, 20-30 μm sized area (indicated by the red arrow) in dark grey, indicat- ing that here the leading edge of the p1 peak is closer to the Fermi level than in other areas and hence that this area is more metallic. Figure 7.1j shows a map based on the p3 position, which by comparison with Figure 7.1g is more sensitive to different domain orientations. Interestingly, the map shows a strong change in the p3 position (indicated by the green arrow in Figure 7.1i) in a region that seems homogeneous in the microscope image (compare with Figure 7.1a). This likely hints towards a kink within the same sur- face layer, where one side is slightly tilting away from the surface. These observations emphasize the complexity of the surface structure of a cleaved crystal, with various dif- ferent steps and domains on various different length scales, which can hardly be fully accessed by a single technique.

Instead of fitting the spectra, contrast based on the near-EF states can be also obtained by integrating the normalized spectral weight close to the Fermi level. Figure 7.1j shows a

map based on Irat as defined previously in Figure 6.9d, i.e. the intensity ratio of the ratio of spectral weight at the Fermi level divided by the spectral weight at the position of the LHB. The resulting map is very similar to the one obtained by fitting the p1 positions in Figure 7.1h. Because peak fitting is more prone to errors and becomes more complicated for the metallic metastable state where technically a Fermi-Dirac distribution function has to be added, hereinafter Irat will be used to map changes in the near EF states in a spatially resolved way.

7.2. Spatial Evolution of the Metastable State after Current Pulsing

7.2.1. First Pulsing Sequence

Figure 7.2.: a) Irat map of the sample after cleaving superimposed to a microscope image. b) Map after applying a 10 ms, 105 mA square shaped pulse along A (direction indicated by red

arrow). c) Map after applying a 10 ms, 105 mA square shaped pulse along B (direction indicated by green arrow). d) Map after applying 3 more pulses along B and then another 10 ms, 105 mA square shaped pulse along A.

Figure 7.2 shows spatial maps of Irat during a first pulsing sequence. Figure 7.2a de- picts the sample right after cleaving, while Figure 7.2b and c show maps after running a 105 mA, 10 ms square-wave current pulse along A (i.e. along the crystallographic axis [010]) from the bottom left to the top right, pulsing direction indicated by the red arrow) and subsequently after running a current pulse with the same parameters through the or- thogonal direction B (i.e. along [21̄0], pulsing direction indicated by green arrow). After three more pulses along B (not shown), another pulse along A is applied (Figure 7.2d). For comparison with the real space structure all Irat maps are superimposed to an image of the sample taken with an optical microscope after the experiment.

After running the first pulse along A (Figure 7.2b), a large metallic area (~56μm width) spanning along the whole pulse bar has formed, indicated by the increased intensity at the Fermi level. Comparison with the microscope image clearly shows how multiple dif- ferent surface planes become metallic. Furthermore, the weakly metallic patch seen in the pristine state after cleaving disappears. The origin of this metallic patch as well as its behavior after current pulsing will be discussed separately in subsection 7.3.1. Running a pulse along B (Figure 7.2c) causes the formation of new metallic regions along that di- rection, however less pronounced than for the first pulse along A. At the same time, the metallic region formed after the first pulse decreases considerably in size and gets par- tially erased. After running three more pulses along B direction (not shown), Figure 7.2d shows a map taken after another pulse along A. It shows that the previously erased area becomes metallic again and that the overall metallic area becomes slightly larger than before. At the same time, parts of the metallic area in the top left corner, formed after the first pulse along B, get erased. Note that between Figure 7.2 c and d all spectra across the whole device surface became broader, resulting in a subtle overall increase of Irat. This effect is discussed in detail in Appendix A.2.2, it is likely caused by a surface contamina- tion triggered by the current pulse and primarily affects the sharpness of the spectra but has no influence of the qualitative behavior.

Overall, these first spatial maps clearly show that running current pulses along orthog- onal directions does not only create metastable metallic regions along that direction, but

 also simultaneously erases previously existing metallic states in adjacent areas.

7.2.2. Spatially Resolved Electronic Structure

Figure 7.3.: a-d) Spatial Irat maps of the pristine, freshly cleaved sample (a) and after run- ning a pulse along A (b), a subsequent pulse along B (c) as well as after three more pulses along B (d). The region on the sample surface where the scans are taken is indicated by the blue box in Figure 7.2. e1-e3) ARPES image plots extracted from the area indicated by the magenta box in panels a-c, respec- tively. A Fourier filter is applied to remove the grid pattern in the spectra caused by the fixed analyzer mode. f1-f2) EDCs extracted at kx=0A˚-1(f1) and 0.3A˚-1(f2), respectively, from spectra taken in the magenta regions indicated in panels a-d. EDCs are smoothed using the Gaussian method over a window of 12.5 meV. g1-g3) Analogous to e1-e3

for the region indicated by the orange box in panels a-c. h1-h2) Analogous to f1-f2 for the region indicated by the orange box in panels a-d.

To understand this intricate interplay of writing and erasing metallic areas better, the electronic structure within individual surface domains will be studied in Figure 7.3, fo- cusing on a small 80×60 μm region of the cleaved surface. Here, in contrast to the larger scans in Figure 7.2, the maps are taken with a smaller stepsize (2μm instead of 3μm), and the sample is locally moved with a piezo scanner instead of through the whole manipu- lator, giving overall higher spatial resolution. Figure 7.3a shows such a fine map of Irat in the freshly cleaved sample, for better orientation the region where the map is taken is indicated by the blue box in Figure 7.2b. Like the coarse map in Figure 7.2a, the fine map also shows the slightly metallic patch mentioned previously which will be discussed sep- arately in subsection 7.3.1. Figure 7.3b shows the same area after running the first pulse along A direction, clearly showing the creation of a metallic region while the metallic

patch seen in the pristine sample disappears. Within individual domains the current created metallic states are homogeneous within the spatial resolution of the experiment, however one can also identify a thin inhomogeneous boundary area (indicated by green arrow). Note that this boundary in the electronic structures coincides with real space boundaries between different surface planes (compare to Figure 7.1a and Figure 7.2b); its nature will also be discussed more in subsection 7.3.1. Figure 7.3c shows the same area after running a pulse along the orthogonal direction (B) which causes a large portion of the metastable metallic state to disappear, while there is still a persisting metallic area in the upper region of the scan window. Running another three pulses along B (Figure 7.3d) further reduces the overall area covered by the metallic state created after the first pulse, shrinking from an extended area in panel c into a small inhomogeneous island, while new metallic areas at the top right corner appear.

Figure 7.3e1-e3 show spectra extracted from the region indicated by the magenta box in the fine scan, which by comparison with the microscope image belong to a single ho- mogeneous surface plane. Note that to ensure a reasonable measurement time, all spatial scans were taken using the so-called fixed acquisition mode, which leads to a gridlike intensity modulation in the ARPES spectra [214]. Therefore, a Fourier filter is applied to remove this pattern from the image plots. Figure 7.3e1 shows the CCDW ground state with the characteristic gapped LHB, albeit not being perfectly oriented along Γ-M di- rection (cut is slightly off in ky). After the first current pulse (Figure 7.3e2), this area transforms into the metastable metallic state. This confirms that the increase of Irat seen in the spatial maps is indeed due to the formation of the metastable state consistent with the spectra in Figure 6.3. Notably, also here a shoulder at the LHB position can be seen in the EDCs (Figure 7.3f2). After running an orthogonal pulse along B, the gap in this area opens again (Figure 7.3e3). However, close inspection of the EDCs (Figure 7.3f1 and f2) reveals that the LHB does not fully go back to its original position and shape but instead the intensity remains slightly suppressed with more spectral weight close to the Fermi level and the leading edge at slightly lower binding energy. After three more pulses along B, the LHB recovers slightly further with the shoulder becoming weaker and the intensity increasing more. This trend is very similar to what was seen in the tempera- ture ramp (Figure 6.9) and thus suggests that the erasing effect could be a thermal effect caused by Joule heating. The creation of heat is furthermore already indicated after run- ning the first pulse along A, where a small region parallel to the metallic area becomes more insulating (Figure 7.3b), reminiscent of an annealing effect. This will be discussed in more detail later in in subsection 7.2.4.

Figure 7.3g1-g3 and h1-h2 show spectra and EDCs, respectively, extracted from the area indicated by the orange boxes on the sample surface. While here, similar to the magenta region, after the first pulse along A the metastable state is created, the spectrum stays metallic after the perpendicular pulse along B direction. Nevertheless, close comparison of the EDCs in Figure 7.3h1-h2 shows that the spectrum does not stay exactly the same but instead becomes slightly less metallic. Note that the X-ray beam itself can also cause a very subtle erasing effect, this effect however is weaker than the changes that are ob- served here after the pulses along A and B (see Appendix A.2.1 for a discussion).

The direct comparison of spectra extracted from the orange and magenta box demon- strates how after the erase pulse regions with different electronic structure are created. These states are reminiscent of the intermediate states seen in the temperature ramp in section 6.5, consistent with a spatially inhomogeneous transient heating effect caused by

the current pulse. However, at elevated temperatures the lifetime of the metastable state decreases exponentially, meaning that the intermediate states seen in Figure 6.9 are only stable for a short period of time. In contrast, no changes in the intermediate states can be observed even after over 35 min. Thus, applying pulses along orthogonal directions allows to create stable states with intermediate electronic structure in between metastable and CCDW state.

7.2.3. Overview over all Pulses

Figure 7.4.: a) I-XIII are Irat maps for multiple current pulsing sequences. Pulsing di- rection and number of pulses are labeled in red and green, respectively. All pulses were square-shaped, had an amplitude of 105 mA and a duration of 10ms. b) Measured resistances along A and B after applying the current pulses. Roman number indicate to which Irat map the transport data cor- responds to.

After introducing the principal effects of running 105 mA, 10 ms square pulses along or- thogonal directions through the sample, Figure 7.4 gives spatial maps of Irat for multiple pulsing sequences applied on the same sample. Pulsing directions along A and B are indicated by the red and green arrows, respectively. Due to time constraints a slightly smaller window was recorded after the first pulses (Figure 7.2). Therefore, for easier comparison the first three panels (I-III) as well as panel V, identical to Figure 7.2a-d, are also shown in this smaller measurement window. For better orientation, the inset in the first panel schematically indicates the location of the measurement window on the device surface.

Visual inspection of Figure 7.4I-XIII reproduces the trend seen in Figure 7.2 consistently over all cycles, namely that pulsing along one direction leads to creation of metallic state along that direction while parts of the pre-existing metallic states get erased. This is more clear for metallic states along A, since here a larger metallic area is created whereas the metallic region created after the pulse along B direction is smaller and initially largely outside of the smaller measurement window (compare with the larger window size in Figure 7.2). Nevertheless, as already observed in Figure 7.2b-d, the erasing process pro- cess is qualitatively similar along both directions and can also be seen after pulses along A direction in the center right area of the maps in Figure 7.4 (indicated by orange arrows).

Running multiple pulses along the same direction makes these effects more pronounced. For instance, compare panel III with IV, where 3 additional pulses along B direction lead to the creation of more metallic regions (top right corner), while the metallic state created after the first pulse along A gets almost completely erased. Note that while applying multiple pulses along the same direction increases the metallic area along that direction, the electronic structure in regions that were already metallic after the first pulse does not show any clear changes, in accordance to what was observed in the microARPES data in the previous chapter.

With ongoing number of sequences of perpendicular pulses the erasing effect appears to become overall weaker and creates a large variety of intermediate states (compare panels II, VI and IX). Nevertheless, applying multiple pulses along B leads to the metallic area along A getting almost fully erased again (panels IX-XII). This furthermore eventually causes the formation of a pronounced metallic region along B (XII), similar in width to the one formed along A.

The overview in Figure 7.4 also shows that regions that turned metallic after a pulse along A and got erased by a perpendicular pulse along B, always turn metallic again after an- other subsequent pulse along A and vice versa. This suggests some form of memory effect and is likely related to the fact that the erased areas do not fully go back into the pristine CCDW state as seen in Figure 7.3f1-f2. Here, the remaining weak shoulder in the EDCs is possible related to some remaining discommensurations or a small amount of metallic areas, which might act as hot spots for the next current pulse along that direc- tion.

The setup of the experiment also allows to directly compare the observed changes in electronic structure on the surface of the device with the bulk transport. Figure 7.4b gives the measured resistances R along A and B, respectively, corresponding to the spec- tra in Figure 7.4I-XIII. It shows that after the initial pulse, pulsing along one direction lowers the R along that direction, while it increases along the orthogonal direction, in accordance with a previous report [163]. When applying multiple pulses along the same direction (for example between pulses IX-XII), R keeps increasing along the orthogonal direction with every pulse while it stays constant along the pulsing direction after the

first pulse. This is in contrast to the ARPES maps, where clearly the metallic area in- creases in size with ongoing number of pulse. This observation can be explained with a recent study [163], which found that the current in the metastable state is mostly carried by a localized conducting channel, often appearing at the device boundary formed along the pulsing direction. Pulsing along the orthogonal direction leads to the formation of a conduction channel along this direction as well while breaking the channel along the original pulsing direction. As such, transport is dominated by this conduction channel and a larger area of metastable state will not lower the resistance any further.

7.2.4. Spatial Profile of the Writing Effect

Figure 7.5.: a) Irat map after applying the first current pulse along A (map identical to Figure 7.2b) b) Difference between Irat map in the pristine CDW state and after the first pulse was applied along A. c-f) Cuts through Irat maps in the pristine, freshly cleaved state (black) and after running the first current pulse along A (orange). The regions where the cuts are taken are indicated in panel a. Dashed grey lines in panel e and f indicate the spatial resolution of the scan as extracted from the edge of the sample.

Figure 7.5 studies the homogeneity of the current-induced metastable state in more de- tail. Figure 7.5a shows a map after the first current pulse (identical to Figure 7.2b), while Figure 7.5b shows a difference map where the Irat map of the freshly cleaved sample is subtracted from the map taken after the first pulse (i.e. (Figure 7.2a is subtracted from Fig- ure 7.2b). Red signals an increase and blue a decrease. This difference map clearly shows a strong increase of Irat across most of the whole pulse bar. It also displays narrow areas with a subtle decrease of Irat, running parallel above and below the newly formed metallic regions. To study these effects in more detail, Figure 7.5c and d show cuts through the Irat maps taken through the area where the metastable state is formed, i.e. where Irat increases (Figure 7.5c, area where the cut is taken is indicated by the green bar in panel a) and the area parallel where Irat decreases (Figure 7.5d, indicated by magenta bar in panel a). The black curve is extracted from the pristine, freshly cleaved state while the orange line is taken from the Irat map after the first pulse along A. Figure 7.5c shows that Irat of the metallic area is quite homogeneous, with less variations than the pristine sample. The

only notable variations are likely artifacts caused by different domain orientations (see Figure 7.1). This indicates that within the errors given by the different domain orienta- tions, the electronic structure of the newly formed metastable state is qualitatively very similar everywhere on the sample surface.

Figure 7.5d displays a cut through the area in which some regions become less metallic after the first current pulse. It shows how in the freshly cleaved area there are regions with strongly varying intensity at the Fermi level, whereas after the pulse Irat decreases to the value of the adjacent areas and becomes homogeneous. This is reminiscent of an annealing process, possibly caused by heat generated by the current pulse that is dis- sipating to the sides. Alternatively, there could be some weaker stray current running through this area that is not sufficient to switch this area into the metastable state.

To analyze the sharpness of the boundary between metastable and CCDW state, Fig- ure 7.5e and f show cuts perpendicular to the pulsing direction. The cuts are chosen by comparison with the optical microscope image to ensure the boundaries are within the same surface layers. For both cuts, the edge of the metastable state appears to be quite sharp. For reference, a dashed grey line indicating the spatial resolution of the scan (10μ as extracted from the edge of the device) is superimposed. This clearly shows that the width of the edge of the metallic state is for both cuts in Figure 7.5e and f within the spa- tial resolution of the experiment. Moreover, Figure

7.5e also shows a subtle annealing effect (indicated by the blue arrow) in close vicinity to the edge of the metastable state. The observation of a sharp edge of the metastable state within the experimental resolu- tion is in accordance with previous STM experiments, which found an atomically sharp boundary after applying voltage pulses through the STM tip [136, 159]. In contrast, no clear indications for the recently reported conduction channels can be found [163]. How- ever, since that data was obtained with a bulk sensitive technique it is not clear if those channels can be seen on the surface, moreover they are likely highly localized on the edge of the sample (below the spatial resolution of the SQUID experiment of 2μm) and would thus require additional studies with improved spatial resolution.
The homogeneity of the metastable state and the well defined boundary to CCDW re- gions indicate sharp threshold conditions for the creation of the metastable state. How- ever, as the switching mechanism is yet not well understood no definitive assignment can be made about which quantities are decisive, since factors such as the energy deposited by the current pulse, the amount of Joule heat generated, but also the heating and cool- ing rate likely play a role. A possible scenario is that areas within the direct current path that get exposed to the highest energy switch directly into the metastable state as long as the local current density is above a certain threshold and generated Joule heat dissipates away quickly enough, whereas the adjacent areas that become less metallic are likely only exposed to heat dissipating away from the center region or alternatively a weaker current density.
However, the situation becomes more complicated at regions where the boundary of the metastable state corresponds to boundaries between different surface layers visible in the optical microscope image. This is clearest in the area indicated by the yellow arrow in Figure 7.5a (and also the green arrow in Figure 7.3b), which shows a region with interme- diate states signaled by the dark grey color. This suggests additional effects introduced by different layers of different height, which might for instance affect the dissipation of the generated Joule heat. This boundary region in particular will be discussed in detail in subsection 7.3.1.
Figure 7.4 shows that while a pronounced metallic bar covering a large fraction of the pulse bar is easily formed after the first pulse, the metallic area along the other direc-

Figure 7.6.: a) Microscope image of another cleaved sample, taken after the experiment. b-c) Irat maps of the pristine, freshly cleaved surface as well as after running three 10 ms, 105 mA current pulses along B.
tion is initially considerably smaller. Since the direction of the first pulse (A) is along the principal axis of the unreconstructed lattice ([010], i.e. Γ-M direction), and the other pulsing direction ((B) is orthogonal (along [21$\bar{1}$0]), not aligned with a principal axis) this could suggest the efficiency or probability of switching into the metastable state is de- pendent on the direction of the current pulse. However, in a different sample shown in Figure 7.6 this effect is not reproduced. Figure 7.6a shows a microscope image of a dif- ferent sample taken after the experiment, Figure 7.6b shows the Irat map of the freshly cleaved sample. In this device, a pronounced metallic region is created along the [21$\bar{1}$0] direction even after the first pulse, and after three more pulses most of the pulse bar turns metallic (Figure 7.6c). This suggests that the asymmetry in metallic regions created after pulsing along A and B is likely due to the sample morphology instead of some

intrinsic directionality related to the principal axes of the lattice. Note that this device also shows the annealing effect of regions parallel to the pulsing direction, where slightly metallic areas turn into more insulating regions (see for instance regions indicated by the blue arrows in Figure 7.6b and c).

7.2.5. Spatial Profile of the Erasing Effect

Figure 7.7.: a1-a3) Irat maps taken after a pulse along B subtracted after from maps taken after a pulse along A. Labels in Roman numbers correspond to the maps in Figure 7.4. Insets show the difference maps after applying 3 more multiple pulses along B. b1-b3) Difference maps from panels a1-a3 superimposed to the Irat maps taken after the previous pulse. For instance, in panel b1 the Irat map after a pulse along A is shown, corresponding to panel II in Fig- ure 7.4. Superimposed is the difference plot where the Irat map corresponding to panel III is subtracted from the one corresponding to II. For clarity the Irat maps are masked to only show the metallic regions. c1-c3) Difference maps from panels a1-a3 superimposed to an optical microscope image taken after the experiment. For clarity all boundaries visible form the microscope image are additionally drawn in grey lines. d1-d3) Difference maps from panels a1- a3 superimposed to the p3 map of the pristine sample.

After analyzing the writing effect, this section focuses on the erasing effect. To better vi- sualize the erasing patterns, Figure 7.7a1-a3 show difference maps, where Irat maps taken after a pulse along B pulse are subtracted from maps taken after a pulse along A. Blue signals a decrease of Irat and thus indicates that the sample becomes less metallic in those regions, red signals an increase due to the formation of new metallic regions. The spatial pattern of the erasing effect looks qualitatively similar for all three sequences. However, the absolute differences becomes overall smaller, indicating a saturation. The insets show difference maps after 3 more pulses are applied along B. Here, regions that did not get erased after the first pulse along B get erased while more metallic areas are created along the pulsing direction.

By inspection of Figure 7.4 and Figure 7.7a1-a3 it appears that the erasing is the strongest at the outer boundaries of the metastable state. To confirm this observation and visual- ize more clearly which parts of the metastable state get erased, in the second row (Fig- ure 7.7b1-b3) the difference spectra are superimposed on the Irat maps taken after the previous pulse. For instance, the difference map obtained by subtracting the map from Figure 7.4II from the map after the subsequent orthogonal pulse in Figure 7.4III, is plot- ted on top of the original Irat map in Figure 7.4II. For clarity, the Irat map is masked to only show the metallic regions in grey. This presentation corroborates the assessment that the metallic regions get most erased at the boundary between CCDW and metastable state. With increasing number of pulses along A direction during the course of the experi- ment, the metallic region becomes slightly larger (compare grey areas in Figure 7.7b1-b3) and the erasing hot spots move accordingly. This behavior, i.e. the erasing being most pronounced at the interface between metastable and CCDW phase, can also be seen af- ter applying multiple pulses along the same direction (insets

Figure 7.7b1 and b3). This causes metallic regions which remained after the first pulse along B to shrink into smaller islands after additional pulse are applied (Figure 7.4IV and X.

To investigate the influence of the microscopic surface structure on the erasing pattern, in Figure 7.7c1-c3 the difference maps are plotted on top of microscope images. For further clarity, all boundaries from the microscope image are drawn in grey lines. By compar- ison, these seem to be mostly correlated with erasing hot spots that run parallel to the pulsing direction (marked by the orange arrows).

As previously shown in Figure 7.1, the surface also possesses domain boundaries within individual surface layers that cannot be directly seen in the microscope image. Fig- ure 7.7d1-d3 show the difference maps superimposed on the map of the p3 position (see Figure 7.1i). This comparison shows that for instance the erasing hot spot in the bottom left corner (marked with an orange arrow) is correlated with a large change of the p3 position. As discussed previously, this indicates different domain orientations, likely be- cause a flake is slightly tilted away from the surface.

In summary, the erasing patterns have roughly grid-like shapes. While the lines run- ning perpendicular to the pulsing direction are primarily along the interface between CCDW and metastable state, the lines in the erasing pattern running parallel are primar- ily connected to boundaries between different domain orientations. In some places the boundary of the metastable state coincides with the space boundaries between different surface layers so that no clear distinction can be made, however the fact that the eras- ing pattern changes with the area of the metallic state with ongoing pulsing sequences or when applying more pulses along the same direction clearly highlights the impor- tance of the CCDW-metastable interface. This observation can be interpreted in differ- ent ways: As these boundaries naturally provide an interface for the CCDW domain to

grow, the metastable state might decay in such a manner even when homogeneously heating the sample. However, this behavior is usually observed on the nanometer, mi- croscopic length scale and it is unlikely that such an effect plays a role on the microm- eter, mesoscopic length scale of this experiment. Furthermore, preliminary experiments on the closely related light-induced metastable state showed that when homogeneously heating the whole sample the metastable state appears to transition in a spatially homo- geneous manner within the spatial resolution. Another, more plausible explanation is that the spatially inhomogeneous erasing pattern is related to the intensity distribution and dynamics of the physical quantity that causes the erasing, likely heat or current. As for the thermal erasing both the absolute temperature and the time the sample stays at that temperature play a role, two scenarios are possible. On the one hand, the CCDW- metastable state interface and also boundaries between different domains in real-space might be more conductive than the metastable state itself and thus get exposed to most of the generated heat (or current). On the other hand, assuming that the erasing is a thermal effect, the dissipation rate could be the decisive factor. Since a metal is a better thermal conductor than an insulator, the generated Joule heat initially dissipates rapidly through the metastable state. Therefore, the center of the metastable regions might have only been at elevated temperatures for a period shorter than the lifetime of the metastable state and thus not get significantly erased. As the CCDW state conducts heat only poorly, the dissipation rate is lower, causing the metastable state to get more erased at the interface area. This argument could also be used to explain the erasing hot spots around real-space boundaries. The implications will be discussed in more detail later on in section 7.4.

Figure 7.8.: Resistances measured along A (oraage) and B (green) as well as the corresponding changes of Irat within the measured surface area for sequences of square-shaped, 10 ms, 105 mA pulses applied along perpendicular directions.

The erasing effect can be also seen in the more spatially integrated microARPES data. Figure 7.8 displays Irat within the measured region in direct relation to changes in trans- port along both pulsing directions for sequences of orthogonal pulses. It shows how R decreases along the pulsing direction while it increases along the orthogonal direction, in excellent agreement with literature [163]. Initially, the Irat also changes with pulsing di- rection, in accordance to the dynamical changes of the metastable state through writing and partial erasing. However, after multiple pulsing sequences the changes get smaller and eventually Irat converges to an almost constant value. This saturation of the erasing effect is similar to what is observed in the nanoARPES data (Figure 7.4 and Figure 7.7). This emphasizes that the dynamics observed in the nanoARPES data are not related to the surface contamination effect causing the bands to broaden slightly after the first cou- ple of pulses (see Appendix A.2.2). It furthermore emphasizes again that the metastable state on the surface is not directly related to the bulk transport properties. Assuming that the erasing effect is thermal caused by Joule heat, the saturation can be qualitatively

explained from the experimental observations. As discussed in Figure 7.3, the erasing does not fully revert back into the pristine CCDW state. This suggests the resistance in this area is smaller than the initial CCDW state and thus less Joule heat is generated after a current pulse, leading to a weaker erasing effect. Nevertheless, this can only serve as a qualitative explanation. As previously discussed in Figure 7.7, the erasing pattern is a complicated convolution of the boundaries of the metastable state and the morphol- ogy of the sample surface, and as the area of the metastable state slightly increases with every pulse its boundaries might coincide with different structural features which might make the erasing process more or less efficient or shift its center outside the measurement window.

7.3. Further Observations

7.3.1. The Metallic Patch

Figure 7.9.: a-b) Irat map of the pristine, freshly cleaved surface (a) as well as after run- ning the first current pulse along A (b). Maps are identical to Figure 7.3a-b. c1-c2) Spectra extracted from the area marked in brown in panel a in the pris- tine state (c1) and after running the first current pulse (c2). A Fourier filter is applied to remove the grid pattern in the spectra caused by the fixed analyzer mode. d1-d2) Analogous to c1-c2 for spectra extracted in the area marked in green in panel a. e) Spectrum integrated within the area marked in blue in panel b after the first current pulse was applied. f1-f2) Second derivative im- ages, corresponding to the spectra in

panel c1 and c2, respectively. For clarity the plots are smoothed using the Gaussian method over a window of 30 meV and 0.005 A˚-1. g1-g2) Second derivative plots corresponding to the spectra in panel d1 and d2, respectively. h) Second derivative plot corresponding to the spectrum in panel h. i) EDCs extracted at kx=0A˚-1from spectra in the brown, green and blue region in the pristine surface as well as after running the first current pulse along A. j) Same as (i) but for EDCs extracted at kx=-0.24A˚-1.

After analyzing various aspects of the current-induced metastable state, this section fo- cuses on the metallic patch observed on the freshly cleaved surface. For this purpose, Figure 7.9a-b show again the fine spatial scans of the pristine sample surface (panel a) as well as after the first current pulse along A (panel b), respectively, which will this time be analyzed with emphasis on the metallic patch. After the first current pulse (Fig- ure 7.9b) the lower half of the previously metallic patch becomes insulating indicated by

the light gray color (i.e. little intensity close to the Fermi level), whereas a boundary re- gion in between form an intermediate state indicated by the dark grey color. Note that this boundary between metallic patch and insulating states in the freshly cleaved surface is aligned with a boundary between two different surface planes as visible from the mi- croscope image. To understand the origin of this metallic patch and its behavior after current pulsing better, Figure 7.9c1 and d1 as well as c2 and d2 show spectra integrated in the brown and green areas before and after pulsing, respectively. For comparison, the metastable state is also shown in Figure 7.9e, extracted from the area marked in blue in Figure 7.9b). The lower row (Figure 7.9f1-h) shows the corresponding second derivative images. While the as-is spectrum of the metallic patch area after cleaving both in the brown and green area looks like a state between CCDW and current-induced metastable state, the second derivate image reveals characteristic differences in the dispersion. No- tably, the spectrum of the metallic patch shows a clear hole-like feature above the LHB which is absent in both the CCDW and the metastable state. As discussed previously, while the current-induced metastable state also shows a band close to the Fermi level, it consistently has a weak electron-like dispersion (compare to (Figure 7.9h). In the brown area, the hole-like band disappears after the first current pulse (Figure 7.9c2 and f2) and the spectrum is highly reminiscent of the CCDW state. In contrast, the spectrum extrated from the green area (Figure 7.9d2 and g2) shows a remaining weak feature close to the Fermi level. Notably, now this near-EF state does not show the clear hole-like dispersion any more but instead the overall spectrum is similar to one of the intermediate states when heating the current-induced metastable state to about 65 K (compare to Figure 6.9). To analyze this more clearly, Figure 7.9i and j show EDCs taken close to Γ point as well as slightly away, respectively (integration regions shown in Figure 7.9e by the red lines). The EDCs of the metallic patch (dashed brown lines) show a clear shoulder at Γ point close to the Fermi level (panel i), which shifts to higher binding energies and closer to the LHB position away from Γ (panel j), in accordance with its hole-like dispersion seen in the second derivative image (Figure 7.9f1). In the green region, this shoulder disappears or becomes considerably weaker after the first current pulse. In contrast, in the brown region the shape of the EDC at Γ point remains almost unchanged, while away from Γ (Figure 7.9j) the shoulder indicated by the orange arrow becomes weaker and moves closer to the Fermi level. Its position is now similar to the one of the metastable state. This further supports the impression that in this area the metallic patch transforms into a partially quenched metastable state.

The hole-like band to the Fermi level can be seen in multiple other regions on the surface of the same sample and can also be observed on various other samples, albeit usually in smaller areas than here. This indicates that such metallic areas are a generic phe- nomenon on cleaved 1T-TaS2 surfaces, however requiring a small beamsize to be clearly observable with ARPES. It is unlikely that the metallic patch is related to the supercooled state [156, 157], since this state is reported to have a similar real-space structure to the NCCDW phase [136]. In contrast, the bandstructure of the metallic patch observed here looks considerably different than the one of the NCCDW phase (compare to Figure 6.7), where notably no hole-like feature close to the Fermi level exists. A possible explana- tion for the metallic patch might be a (self)doping effect, possibly by excess sulfur atoms, which get evaporated from the surface by the first current pulse. However, no chem- ical potential shifts are observed and the LHB stays at the same binding energy even after the hole-like feature disappears after the first current pulse. The overall most likely reason for the metallic patch are local changes in the layer stacking, which is known to

alter the electonic structure and has been observed in previous experiments [149, 215, 216]. Indeed, the EDCs near Γ point show reminiscence to the previously published STS and angle-integrated photoemission data. The fact that a large fraction of the metallic patch gets erased after the first current pulse indicates that such configurations might also be metastable. Furthermore, the data also suggests that some parts of the metallic patch get converted into a partially quenched current-induced metastable state, hinting towards an intriguing interplay of different metastable configurations. The fact that the metallic patch does not get converted into the full current-induced metastable state (Fig- ure 7.9e) might be related to the boundary between different surface layers of different height. In particular, it is possible that the metallic patch switched into a current-induced metastable state, however the physical inhomogeneities might slow down the dissipation of heat in this area and thus cause its partial erasing right after its creation. Here further studies are necessary to investigate the origin and properties of the metallic patch as well as its interplay with current- or also light- induced metastable state in more detail.

Chapter 7 Figure Details

Figure 7.1

This figure introduces the spatial-mapping workflow by putting microscope images, photoemission intensity maps, ARPES spectra, and derived spatial maps side by side. The overall visual message is that the cleaved surface is highly structured in real space, and those real-space variations show up directly in the measured electronic spectra.

Panel **a** is an optical microscope image of a cleaved device surface. The sample occupies the center and looks like a pale, irregular flake surrounded by darker regions and contact structures near the edges. Several thin cracks, steps, or domain boundaries are visible as faint lines across the surface. Three blue arrows point to notable internal boundaries or step-like features. A white scale bar near the bottom indicates **30 μm**. The main impression is of a nonuniform, multi-domain surface rather than a perfectly flat single terrace.

Panel **b** is a spatial map of integrated photoemission intensity, shown in false color and cropped to the shape of the measured area. The map is mostly brown-to-yellow with patches of green and a pixelated edge. A vertical color bar labeled **I_tot [a.u.]** runs from low to high. Three blue arrows point to features in the map that visually line up with structures in the microscope image. Three small colored squares mark specific regions selected for later spectral extraction: a magenta square near the lower left, an orange square toward the upper middle-left, and a black square near the upper right.

Panel **c** overlays the photoemission map on the optical microscope image. The same irregular sample outline appears, now with the intensity map rendered semi-transparently over the microscope texture. The correspondence between real-space structures and photoemission intensity is made explicit. Black arrows and crystallographic labels **[210]** and **[010]** indicate orientation.

Panels **d**, **e**, and **f** are three ARPES spectra taken from the colored regions marked in panel b. Each is plotted as intensity versus emission angle **φ [deg]** on the horizontal axis and binding energy **E − E_F [eV]** on the vertical axis. The three spectra have similar general structure but differ in the exact horizontal position and curvature of the bright features, which visually conveys that different real-space regions correspond to slightly different cut orientations in momentum space.

Panel **g** is a line plot of angle-integrated spectra for the three regions. Three curves, colored to match the extraction boxes, show three prominent peaks labeled **p1**, **p2**, and **p3**. The x-axis is **E − E_F [eV]** and the y-axis is intensity. The first peak near the Fermi level is the most prominent, while lower-energy peaks form weaker shoulders. The figure text explains that p1 varies little across domains, whereas p3 shifts more strongly.

Panels **h**, **i**, and **j** are grayscale spatial maps derived from the spectra. Panel **h** maps the fitted **p1** position, panel **i** maps the **p3** position, and panel **j** maps **I_rat**. In **h**, a dark patch indicated by a red arrow marks a 20–30 μm region that appears more metallic. In **i**, a green arrow points to a region with a strong p3 shift, suggesting a domain-orientation change or kink not obvious in the microscope image. Panel **j** resembles panel h and is presented as the preferred near-Fermi-level contrast map for later figures. Altogether, the figure shows that structural domains, tilts, and steps on the cleaved surface directly affect the apparent ARPES spectra and derived metallicity maps.

Figure 7.2

This figure shows a sequence of four **I_rat** maps overlaid on the microscope image of the same sample, illustrating how current pulses written along two orthogonal directions create and partially erase metallic regions on the surface. The visual emphasis is on anisotropic, history-dependent spatial switching.

Panel **a** shows the freshly cleaved state. The base microscope image is visible beneath a semi-transparent grayscale intensity map. The sample has an irregular polygon-like outline with contact structures around the edges. The **I_rat** overlay is relatively light and mottled, indicating modest variation but no large continuous metallic bar. A faint patch is present in one region, corresponding to the weakly metallic patch discussed later.

Panel **b** shows the state after a **10 ms, 105 mA square pulse along A**, indicated by a red arrow. A broad dark band now runs diagonally across the device in the same general direction as the pulse path. Because darker values correspond to higher **I_rat** here, this band represents a newly created metastable metallic region spanning much of the pulse bar. The band cuts across multiple surface terraces rather than staying confined to a single domain.

Panel **c** shows the state after a subsequent pulse along the orthogonal direction **B**, indicated by a green arrow. A new dark region appears in the B direction, but it is smaller and less continuous than the metallic bar created by the first A pulse. At the same time, much of the original dark A-oriented bar becomes lighter, showing partial erasure. The map therefore visualizes simultaneous writing in one direction and erasing in the other.

Panel **d** shows the state after three more pulses along **B** followed by another pulse along **A**. The main A-oriented metallic band becomes dark again and slightly broader than before, indicating re-formation of the metastable state along A. Meanwhile, part of the B-induced metallic area in the upper portion becomes lighter, showing reciprocal erasing. The sequence as a whole conveys that pulsing along one axis writes metallicity preferentially along that axis while partially removing metallicity previously created along the orthogonal axis.

Figure 7.3

This figure zooms into a smaller measurement window and combines fine **I_rat** maps with local ARPES spectra and EDCs. The main purpose is to show, at higher spatial resolution, that different microscopic regions respond differently to the write/erase pulse sequence and can end up in distinct intermediate electronic states.

Panels **a–d** are four fine grayscale **I_rat** maps of a small rectangular region. Panel **a** is the pristine state and contains a slightly darker metallic patch. Panel **b**, after the first pulse along A, shows a large dark metallic area occupying much of the center-left region, while the original patch disappears. A thin inhomogeneous boundary region remains, marked by an arrow in the thesis text. Panel **c**, after a pulse along B, shows much of that metallic area becoming lighter again, with only a remnant dark region surviving near the upper part of the window. Panel **d**, after three more pulses along B, shows that the previously extended metallic region has shrunk into a small irregular island, while other parts of the window have changed differently.

Panels **e1–e3** are ARPES image plots from the **magenta-box** region. In the pristine state (**e1**), the spectrum shows a gapped ground-state appearance, with spectral weight concentrated below the Fermi level. After the first A pulse (**e2**), the spectrum becomes clearly more metallic, with increased near-Fermi-level weight and a modified lineshape. After the orthogonal B pulse (**e3**), the gap reopens somewhat, but the spectrum does not return perfectly to its original shape. The corresponding **f1–f2** EDCs compare line cuts at fixed momenta and show that the erased state is only partially restored, consistent with an intermediate rather than fully pristine state.

Panels **g1–g3** are analogous ARPES plots from the **orange-box** region. Here too the first A pulse creates a more metallic spectrum, but after the orthogonal B pulse the spectrum remains more metallic than in the magenta region. The related **h1–h2** EDCs show only partial weakening of metallicity rather than full erasure. The visual contrast between the magenta and orange regions is the key point: different real-space areas of the same sample do not all follow the same write/erase trajectory. Some revert more strongly, while others remain in a partially metallic intermediate state.

Figure 7.4

This figure is a large overview of repeated pulsing cycles. It presents a sequence of **I–XIII** maps along with a resistance plot, making clear that the writing/erasing pattern is reproducible, history-dependent, and only partly mirrored by the bulk transport response.

Part **a** is a grid of thirteen grayscale **I_rat** maps labeled with Roman numerals **I** through **XIII**. The first panel includes an inset showing where the smaller measurement window sits inside the full sample area. Red and green arrows label the pulse direction for each step; additional labels such as **x1**, **x3**, or **x5** indicate how many pulses were applied along that direction. Across the sequence, dark elongated metallic regions repeatedly grow along the direction of the latest pulse, while regions created earlier along the orthogonal direction shrink or break apart. In some panels the metallic region is a broad diagonal band; in others it contracts into isolated dark islands. The repeated alternation between A and B produces a kind of spatial memory pattern rather than a simple reset.

Part **b** is a line graph of resistance **R [Ω]** versus pulse sequence index. An orange curve gives resistance along **A** and a green curve gives resistance along **B**. Below the axis, small red and green arrows repeat the sequence of pulse directions. The two resistance curves tend to move oppositely: after a pulse along one direction, the resistance along that direction drops while the orthogonal-direction resistance rises. Compared with the maps above, the graph shows that the surface metallic area can keep growing visually even when the resistance along the pulsed direction has already saturated, implying that transport is dominated by more localized conduction paths than the full spatial extent of the metallic regions would suggest.

Figure 7.5

This figure focuses on the **writing effect** of the first pulse along A. It combines a post-pulse map, a difference map, and several one-dimensional cuts to show that the written metallic band is spatially homogeneous across its interior, sharp at its edges, and accompanied by weaker neighboring regions that become less metallic.

Panel **a** shows the same post-first-pulse **I_rat** map seen earlier, with several colored bars drawn across it to indicate where line cuts are taken. A green diagonal bar crosses the main dark metallic band. A magenta bar lies nearby in a region that does not become metallic. Short red, blue, and yellow bars indicate transverse cuts or special boundary locations.

Panel **b** is a red-blue difference map, obtained by subtracting the pristine map from the post-pulse map. Red marks an increase in **I_rat** and blue a decrease. The central pulse bar shows a broad red diagonal stripe where metallicity increased strongly. Flanking that stripe are narrower blue regions running roughly parallel to it, indicating locations where the sample became less metallic after the pulse. This is the “annealing-like” side effect discussed in the text.

Panels **c** and **d** are line cuts taken along the diagonal direction. In **c**, the orange post-pulse curve sits well above the black pristine curve over a long region, showing that the written metallic band is broad and fairly uniform. In **d**, the orange curve lies below the black curve, showing that some neighboring regions actually lose near-Fermi-level intensity and become more homogeneous after the pulse.

Panels **e** and **f** are cuts perpendicular to the pulse direction. The transitions between the metallic band and the surrounding regions are steep. Dashed gray lines mark the scan’s estimated spatial resolution. The observed edge width is comparable to that resolution, which visually supports the claim that the real boundary is very sharp. One of the cuts also shows a small dip just outside the main band, marked by a blue arrow, corresponding to a weak local annealing effect adjacent to the metallic region.

Figure 7.6

This figure provides a comparison sample. It shows that the asymmetry between writing along A and along B in the earlier sample is not universal, suggesting that morphology matters more than crystallographic direction alone.

Panel **a** is a microscope image of a different cleaved sample. The device surface fills the center, while dark contact structures sit around the edges. Red arrows mark the crystallographic directions **[210]** and **[010]**. The surface looks smoother in some places and more fractured in others, but its overall morphology differs from the earlier sample.

Panel **b** is the pristine **I_rat** map of this new sample. It is a grayscale map with modest contrast and some lighter and darker patches. A blue arrow marks a slightly metallic region. A **40 µm** scale bar is shown.

Panel **c** is the map after **three pulses along B**. A large dark band now occupies much of the right half of the map, aligned approximately with the B direction. This metallic region is much more pronounced than the small B-induced metallic area in the previous sample. A green arrow labeled **B x3** points along the written direction. A blue arrow again marks a neighboring region that appears to have become less metallic. The comparison with panel b makes the writing effect along B visually obvious. The figure's key message is that strong B-direction writing is possible and therefore the earlier A/B asymmetry is probably not intrinsic.

Figure 7.7

This figure dissects the **erasing effect** using difference maps. It asks where metallicity is lost after pulsing along the orthogonal direction and how those erasing hot spots relate to the pre-existing metallic regions, the microscope-visible boundaries, and the p3 domain-orientation map.

The top row, **a1–a3**, contains blue-red difference maps labeled as differences between maps taken after pulsing along A and the subsequent pulse along B. Blue denotes decreased **I_rat** and therefore erasure of metallicity; red denotes increased **I_rat** and new writing. In each panel, the dominant feature is a blue, somewhat grid-like pattern. Insets show analogous differences after three more B pulses, where remaining metallic fragments are erased further and some new B-oriented metallicity appears.

The second row, **b1–b3**, overlays those difference maps on the metallic regions from the preceding state. The metallic regions are shown in gray, and the blue erasing pattern sits mostly at their boundaries. This makes the main geometric point visually clear: erasing is strongest not in the middle of large metallic bars but near their outer edges and interfaces with surrounding phases.

The third row, **c1–c3**, places the difference maps on top of the optical microscope image. Gray lines trace visible structural boundaries from the microscope view, and orange arrows point to places where the erasing pattern tracks boundaries that run roughly parallel to the pulsing direction. These overlays suggest that sample morphology influences where erasing is strongest.

The bottom row, **d1–d3**, overlays the same difference maps on the **p3** spatial map from Figure 7.1. Here the grayscale background encodes p3 position, which is sensitive to domain orientation. Orange arrows point to erasing hot spots that line up with strong p3 changes, meaning that not all important boundaries are visible in the microscope image; some correspond to subtle tilts or kinks within a surface layer. The total visual impression is of a grid-like erasing pattern built from two components: one associated with the metastable–CCDW interface, and another associated with structural or orientational domain boundaries.

Figure 7.8

This is a compact summary plot that compares transport and surface metallicity through repeated orthogonal pulsing. It visually compresses the behavior of the larger mapping experiments into one graph.

The left y-axis gives **R [Ω]**. Two solid step-like curves are shown: orange for **R along A** and green for **R along B**. The right y-axis, in blue, gives **I_rat [-]**. A dashed blue curve tracks the average or integrated near-Fermi-level spectral weight in the measured region. Along the bottom, alternating red and green arrows indicate the sequence of pulse directions.

After each pulse, the resistance along the pulsed direction decreases while the orthogonal one increases, producing an alternating staircase pattern in orange and green. The dashed blue **I_rat** curve initially oscillates as writing and partial erasing proceed, then gradually settles toward a roughly constant level. The visual takeaway is that the surface-sensitive ARPES metric saturates while the two transport channels continue to show directional contrast, reinforcing the idea that surface metallic area and bulk/device transport are related but not equivalent observables.

Figure 7.9

This figure analyzes the **metallic patch** seen on the pristine surface before pulsing. It compares local spectra from that patch before and after the first pulse with spectra from the fully written metastable state, using raw ARPES maps, second-derivative images, and EDC comparisons. The main visual conclusion is that the metallic patch is not simply the same as the current-induced metastable state.

Panels **a** and **b** are fine grayscale **I_rat** maps of the same small region before and after the first pulse along A. In **a**, two colored polygons mark a **brown area** and **green area** within the original metallic patch. In **b**, a blue rectangle marks a region that has become part of the pulse-induced metastable metallic state. After the pulse, the lower portion of the original patch becomes lighter and therefore less metallic, while a darker intermediate boundary remains between regions.

Panels **c1–c2** are ARPES spectra from the brown area before and after the pulse. Before pulsing (**c1**) the spectrum has a relatively unusual near-Fermi-level structure. After pulsing (**c2**) it looks more like an insulating CCDW-like spectrum. Panels **d1–d2** do the same for the green area; after pulsing, this region still retains some near-Fermi-level weight and looks more intermediate. Panel **e** is a spectrum from the blue post-pulse region that represents the conventional current-induced metastable metallic state.

Panels **f1–f2**, **g1–g2**, and **h** are second-derivative versions of those spectra, which make the dispersions easier to see. The pristine metallic patch shows a **hole-like** feature above the lower Hubbard band that is absent from both the ordinary CCDW state and the current-induced metastable state. After pulsing, that hole-like feature disappears in the brown region and weakens or changes character in the green region.

Panels **i** and **j** are EDC comparisons at two momentum values, **$k_x = 0$ Å^{-1}** and **$k_x = -0.24$ Å^{-1}**. Multiple colored curves compare the brown, green, and blue regions before and after pulsing; dashed lines denote pristine spectra and solid lines denote post-pulse spectra. At Γ, the metallic patch shows a shoulder close to the Fermi level. Away from Γ, that shoulder shifts in a manner consistent with a hole-like band. After the first pulse, the shoulder weakens or moves, and in some regions becomes more similar to the metastable-state spectrum. Visually, the figure supports the interpretation that the pristine metallic patch is a distinct electronic configuration, perhaps tied to local stacking variation, and that the first current pulse partly erases it and partly converts portions of it into an intermediate or partially quenched metastable state.

The two conceptual slides at the beginning

The first text-only slide, titled **"Ideas on how to use ML/AI,"** is a brainstorming slide. It has a plain light-gray background with a numbered list of four proposals: infer local surface orientation, map each ARPES spectrum to a true Γ–M direction, use spectral unmixing to enhance faint features and effective spatial resolution, and analyze temperature ramps for intermediate states. There are no graphics; the emphasis is on the research program and what AI would help disentangle.

The second conceptual slide is a workflow diagram. On the left are two small ARPES-like spectra and a real-space sample map with highlighted regions; in the center is a large **AI** label with a two-headed arrow; on the right is a 3D spectral cube labeled in **E**, **θ**, and **φ**. Underneath are "Main goals" and "Further goals," including correction of structural variations, detection of subtle electronic features, mapping of domain alignment and boundaries, automatic alignment, selection of interesting sample regions, and spectral unmixing. The visual message is that AI should connect real-space heterogeneity and full spectral data cubes in both directions.

The third conceptual slide is a schematic about the **erasing pattern**. It shows a simple black rectangular shape on the left, then a colored version on the right annotated with different boundary types: ground-state/metastable-state boundary, boundary between two surface layers, a flake-bending boundary, and a layer-termination boundary. Large text asks whether erased regions correspond to real-space boundaries or electronic boundaries and notes that structural-domain artifacts make the interpretation difficult. It is not raw data, but rather a conceptual map of the hypotheses behind the erasing analysis shown later in Figures 7.7 and 7.8.

The fourth conceptual slide, titled **"Spectral Weight Transfer,"** shows temperature-dependent ARPES band maps across the top, a set of EDCs at fixed momentum, and real-space phase maps across the bottom transitioning from blue insulating regions to red metallic regions with increasing temperature. The adjacent text asks whether the observed spectra are just mixtures of ground and metastable states or whether more exotic intermediate pseudogapped states are hidden in the data. This slide visually motivates later use of spectral decomposition and comparison to thermal intermediate states.

Max Meeting Notes

Max Meeting (3.12.2026)

- Corrected data for Fermi level
- Nano-ARPES tricky as use reference for that dataset, not perfect but a lot of variations in beam times, nano-ARPES data
- Spatial scans in x,y, energy and momentum
- How to process data?
- Current pulses, looking at boundaries (NMF techniques nice, on cleaved sample have a lot of inhomogeneities, create more inhume….) NMF convolutes everything, hard to extract the geometric information
- Some regions look metallic, but because less intense at Fermi level
- Current pulse runs along orthogonal directions, hard to say why this is happening; if there's any logic behind it, happening most along boundaries, different planes and boundary of plane happening on layer of crystal or is it an electronic boundary?
- If could understand these patterns, how far we'd get from that?
- Hirotaka does classification models
- What's happening around the current is phase changes, dissipation of homogeneity; there is some currently unknown effect occurring around the current itself; if those changes can be predicted by the state around the current, see if ML can find correlation of something
- What are some things it could be correlated to?
    - Interesting to know if insulating state, move into metastable metallic state; is there one metastable state, or slightly different ones? Is there a spatial variation effecting the electronic structure, if switch back into insulating state
        - Can we understand the different structures, gradient, cutoff of spatial dimension; understand how the erasing
    - Shooting at other direction, erasing at seemingly random spots, but could find correlation between how spots erase and if they correlate to other levels of metastable states
    - Can we classify all metastable states in the sample? Does current take specific direction through these states? Several different colors before he ran current through it, could be topological differences in current
    - Find some other correlation in after state and erasure; what states get turned into the more metallic state, which states revert back to insulating state?
    - Looking at bridge shape and edge, what did he measure and what was strange about it?
    - Other thing: talking about mapping, Brillioun zone maps across entire device, see if can correlate differences in Brillioun zone with how band structure looks like; if find mapping between the two (how Brillioun zone looks at one location versus another), could normalize everything such that band structures are corrected for topological differences in the device

- Talk to Summer about what she's done and the data she has, see what exactly we have to do to measure and correct stuff
- Max gave corrected data, all files for everything we need
- How to find correlation between state of sample before and after current is passed through?
- Sample has many x y points, many different states before even start (topological differences, not completely flat, has inhomogeneity); looking at different samples on one sample
- Assume current passes through one state instead of many states
- Classify all regions with states beginning and after; are there certain states before that are effected by the current, current takes path through the sample (electron don't flow completely homogeneously through the sample)
- When run current through opposite direction, metallic states, can we classify these states are confidently said to be insulating or states confidently said to be erased; these are states that will be turned this way or that way; current affected by previously metallic or insulating state
- If different colors popping up, set boundaries for what different metastable states could be
- Normalizing for the slopes (correct for that), could use that corrected data on seeing what happens between states

Max Meeting (11.27.2025)

- Made a device looking like a cross, make current across cross and measure with ARPES; when run current pulse, has metastable state that's really long-lived; measure with a normal equilibrium measurement, measured at the beam line, have a small spot size (can scan whole surface of device, map the device with photoemission and changes in electronic structure)
- Map device to microscope image, can see characteristic things with the blue arrows
- Metastable state is metallic, so write new metallic regions and erase old metallic regions
- Is there anything in electronic structure telling us if metallic or erased? Confident not any dependence on electronic structure
- Make code reading out structures of electronic domains and see if there's a correlation between the electronic domains
- Have full maps of energy and momentum, lots of information in the band structure, more sophisticated analysis can do but hard to do in these different domains because gap sizes get too big
- If have code that can look for where is in Brilloun zone, can correct for effects and get clean image of gap size and can look at every single pixel
- Run current pulse through isolating sample, create a lot of heat when running orthogonal current pulse where heat dissipated very efficiently
  - If goes to 50-60 K, goes back to insulating state
- Used for COMSOL very numerical software to simulate heat, but not very accurate (got understanding of fundamental effects but not useful for proving there is a heating effect); current level flows homogeneous or not homogeneously in the sample
- Need more data to train model
- Effects see are convolution of electronic structure and other differences to get rid of electronic differences
- Focus on data analysis to understand the patterns created in the material; best to check if there's something in electronic structure to make it metallic or less metallic

        - If rotational alignment with current with respect to Brilloin zone along different axis (material has a chirality), forms Stars of David that align in a commensurate order; have mirror symmetry (opposite form but have same energy), must have full scan in momentum space; if code is really good, could extract some small features and get chirality of the domains
    - Can use AI to make stronger point? Code to find something interesting like a correlation (like a clustering technique)?
    - Doing ML in code, when he graduated now Lanzara cares
    - Plot structure boundaries - say erasing is correlated to the boundary, then can say the boundary is more conductive; plot different domains in a map
    - Different spectra are different cuts in k-space, assign each cut to a color and plot in a map that tells us a clustering

Max AI Slides

Yes. Here is a careful transcription of each slide, along with a detailed description of the visuals.

Slide 1

Title

Ideas on how to use ML/AI

Transcription

1. **Use it to extract the orientation of each point on the sample surface and then plot maps as a function of kx or ky (and maybe the chirality, alpha or beta); this way we could be a better picture of different kinds of domain boundaries**
2. **use that knowledge to map each ARPES spectrum of the sample surface to the true Gamma-M direction, so we can analyze the electronic structure free of the structural artefacts (or vice versa, we could extract something like a gap size from each spectrum and then find a correction factor to see how large that gap should be at Gamma-M direction) -> This way one could really disentangle which effects come from the electronic structure itself and which are related to the real space structure of the surface; is there anything in the electronic structure that correlates with the probability that it will turn metallic or that the metastable state will get erased. Also there might be subtle differences in the electronic structure between different regions that cannot be identified because there is so many different structural domains (one question for instance is if the metastable state is perfectly homogeneous or if the electronic structure varies)**

3. **Use Conrads spectral unmixing code to enhance spatial resolution of the scans and extract faint features from the band structure that would not be visible otherwise (for instance some weak shadow bands)**
4. **Look into the temperature ramp and identify intermediate states**

Image / layout description

- The slide has a plain light-gray background.
- Large black title at the top left.
- Beneath it is a numbered list of four ideas in black text.
- No plots or diagrams appear on this slide.
- The slide is text-heavy and reads like a brainstorming or planning slide about possible machine-learning applications to ARPES and domain-structure analysis.

Slide 2

Main visible text

Main goals

- **correct for structural variations to extract e.g. self-energy free of artefacts**
- **detect subtle features in the electronic structure**
- **map domain alignment and different types of boundaries**

Further goals

- **automatic alignment**
- **find the most interesting regions on the sample**
- **enhance resolution by spectral unmixing**

Other visible labels and axis text

- Two small spectra on the left each have axes labeled:
  - **E [eV]** on the vertical axis
  - **φ [deg]** on the horizontal axis
- The central sample map has a colorbar labeled:
  - **I_tot [a.u.]**
  - with **low** at the bottom and **high** at the top
- The 3D spectral cube on the right has axes labeled:
  - **E [eV]**
  - **θ [deg]**
  - **φ [deg]**

- The cube’s colorbar is labeled:
  - **I [a.u.]**
  - with **low** at the bottom and **high** at the top
- In the center there is large text:
  - **AI**

Image / figure description

This slide is a conceptual diagram.

Left side

- Two small ARPES-like spectral images are stacked vertically on the far left.
- Each shows a blue-background intensity map with brighter white/red features near the top-middle.
- The top small spectrum has a magenta arrow pointing from it toward a highlighted rectangular region on a central sample map.
- The bottom small spectrum has a black arrow pointing from a dark rectangular region on the sample map toward the spectrum.

Center-left sample map

- There is a roughly square spatial map of a sample, colored in earthy tones: tan, green, brown, and pale yellow.
- The map looks patchy and heterogeneous.
- Several black hand-drawn boundary lines or arrows surround it, suggesting domain boundaries or edge directions.
- A bright magenta rectangular box highlights one region near the upper right of the map.
- A black rectangular box highlights a region in the lower left.

Center

- Between the sample map and the 3D cube is a large **AI** label.
- A thick horizontal double-headed arrow goes left-right through the AI label, indicating two-way mapping between real-space sample data and higher-dimensional spectral data.

Right side

- A 3D blue spectral cube is shown, with bright red and white intensity patches on its top and side faces.
- It looks like a volumetric ARPES or momentum-energy dataset.
- The cube’s axes indicate energy and angular coordinates.

Overall meaning conveyed visually

- The slide is illustrating an AI-driven mapping between local real-space sample regions and high-dimensional spectral information.
- The left side suggests picking particular real-space regions and comparing their spectra.
- The right side suggests reconstructing or interpreting the full spectral data cube.
- The listed goals emphasize correcting structural artifacts, identifying subtle electronic features, boundary mapping, alignment, region prioritization, and spectral unmixing.

Slide 3

Transcription

Top text:

**it would be nice if we could figure out what these regions correspond to,
there are roughly two options:**

- **boundaries between different real space domains, i.e. different flakes or flakes with different tilts**
- **boundaries between different electronic domains, i.e. regions that are metallic and regions that are insulating or differ in their electronic structure in a different way**

=> It would be nice to see if the erased regions correspond to boundaries and if yes, to what kind

Left-middle text:

this is the erasing pattern that we see, it looks roughly like a grid. But why does it look like that? why do exactly these regions get erased while others stay almost the same?

Bottom-left text:

as you can read in chapter 7.2.5 of my thesis the erasing seems to be strongest at the ground state-metastable state boundary but it is hard to make this point with all the structural boundaries

Colored labels on the schematic at right:

- **boundary ground state- metastable state** (red)
- **boundary between two surface layers** (green)
- **boundary between two TaS2 flakes the bend into two different directions** (blue)
- **layer ends here (one layer is higher than the other)** (purple)

Bottom-right text:

a problem here is one to detect real space boundaries (either extract it from the microscope image or orientation of the ARPES cuts), but then these real space boundaries can make it challenging to see if there are any differences in electronic structure because the different domain orientations induce artefacts

Image / figure description

This slide mixes explanatory text with a schematic cartoon.

Left side

- A simple black-outlined tilted rectangular shape is drawn, with one interior dividing line.
- It looks like a block or strip split into sections.
- A gray arrow points down toward this shape from the explanatory text about the erasing pattern.
- This black shape appears to represent the observed erasing pattern, described as grid-like.

Center

- A large gray arrow points from the black-outlined simple rectangle on the left toward a colored annotated version on the right.

Right side schematic

- The same tilted rectangular object is redrawn, now with colored boundary segments:
    - A long red top and bottom outline labeled as the **boundary ground state-metastable state**
    - A blue short segment on the left end labeled as the boundary between two TaS2 flakes bent in different directions
    - A purple interior vertical segment labeled **layer ends here (one layer is higher than the other)**
    - A green short segment on the right end labeled **boundary between two surface layers**
- The intent is to distinguish several physically different types of boundaries that may underlie the same observed grid-like erasing pattern.

Overall message of the slide

- The central scientific question is whether erased regions align with structural boundaries, electronic boundaries, or both.
- The slide proposes that the observed pattern could arise from multiple overlapping boundary types.
- It also emphasizes that structural variation can create artifacts that obscure genuine electronic differences.

Slide 4

Title

Spectral Weight Transfer

Transcription

Top row labels above the band maps:

- **12 K**
- **66 K**
- **70 K**
- **74 K**
- **96 K**

Axes on those panels:

- Vertical axis: **E-E_F [eV]**
- Horizontal axis: **k_x [1/Å]**

Line-plot panel labels:

- **k_x = 0 Å^-1**
- Vertical axis: **intensity [a.u.]**
- Horizontal axis: **E-E_F [eV]**
- Colorbar beside it:
  - **T [K]**

Right-side paragraph:
the question is, do our spectra just show a superposition of ground and metastable regions or are there any exotic intermediate states with pseudogap hidden in there? (see e.g. https://www.science.org/doi/10.1126/science.1150124) => I already tried simple NMF and could only really see two contributions)

Left-side paragraph:
the spectra that we see are likely a superposition of different regions like this here:

it is very likely that there are some regions that have a really interesting electronic structure, but it is hard to identify because of the contribution of the other phases

Bottom row labels above images:

- **T = 164 K**
- **171 K**
- **174 K**
- **177 K**
- **300 K**

Small labels on bottom images:

- first image: **AFI**
- last image: **PM**

Bottom annotations:

- **insulating**
- **metallic**

Bottom center footer:
Thesis Defense
and slide number **4**

Image / figure description

Top-left sequence of spectra

- Five narrow rectangular intensity maps are arranged horizontally.
- Each shows ARPES-like band structure evolving with temperature from 12 K to 96 K.
- Dark blue spectral features are strongest near the top of each panel, just below a dashed horizontal line marking approximately zero energy.
- The spectral shapes appear to broaden and change subtly as temperature increases.

Top-right line plot

- A line plot compares energy-distribution curves at **k_x = 0 Å^-1** for different temperatures.
- The curves are colored according to temperature, with a vertical colorbar labeled **T [K]**.
- The peak position and shape shift across temperatures, illustrating spectral weight transfer.

Bottom row of real-space images

- Five square images are shown from left to right with temperatures 164 K, 171 K, 174 K, 177 K, and 300 K.
- They are false-color maps in blue, white, and red.
- The first image at 164 K is mostly blue and labeled **AFI**, suggesting an insulating phase.
- The final image at 300 K is mostly red and labeled **PM**, suggesting a metallic or paramagnetic phase.

- Intermediate images show mixed red/blue textures with irregular domain-like regions.
- Two black arrows beneath these images point upward toward mixed red/blue regions, indicating coexistence or phase separation.
- The words **insulating** and **metallic** identify the blue and red sides of the phase evolution.

Overall message of the slide

- The slide is about whether measured spectra are just mixtures of known phases or whether truly distinct intermediate states exist.
- The top data show temperature-dependent spectral evolution.
- The bottom maps show a temperature-driven spatial evolution between insulating and metallic regions.
- The mention of NMF suggests the speaker tried simple nonnegative matrix factorization and found only two dominant components, leaving open the question of more subtle hidden states.

If you want, I can also turn this into a cleaner **speaker-notes style transcript** or a **Markdown document with section headings and figure captions**.